

CMPUT 291 Mini Project 2: Furniture Database System using MongoDB

System Overview

This project implements a furniture management command-line system using Python and MongoDB, a NoSQL database. The system manages IKEA furniture items, allowing users to search, check discounts, browse by category and add new items to the database. Python is used mainly as the host language, interacted with command line arguments and menu/control interfaces with MongoDB Query language used to interact directly with the database.

The project's development splits into the two following phases.

- Phase 1 - Build the document database
 - This phase constructs a MongoDB database populated with data from a JSON file.
 - The program takes the JSON filename and MongoDB port number as command-line inputs, connects to the server, and creates a database named 291db with a collection called furniture.
- Phase 2 - Document Interactions
 - This phase allows users to interact with the database through a menu-driven interface.
 - The program connects to the MongoDB server using a specified port and executes queries to search for items, check discounts, browse by category, and add new furniture items to the collection.

User Guide

Prior to running the application, users are first required to set up the 291 database. This is done through running the following command:

```
python load_json.py data/sample_2000.json 27017
```

*The json file path and port number can be replaced with user-specific values.

Upon successfully loading data to the database, to access the full capabilities of the application, the user must run the following command, with the portnumber replaced with the portnumber connecting to MongoDB.

```
python operations.py 27017
```

From there, a menu will appear prompting the user to choose from the following options.

- **Discount Check:** The user is prompted to enter the exact name of a furniture item, and if it exists, the discount status will be displayed. When multiple items share the same name, the user can choose to check one using the unique item ID.
- **Keyword Search:** The user enters a case-sensitive keyword to search the furniture collection for. Matching items are displayed along with their name, category, price, and a short description.
- **Category Search:** The user chooses a category from the present list and all items belonging to said category are displayed.
- **Add Furniture:** Add a new item to the collection, with users prompted to enter the mandatory fields of item ID, name, price, category, a short description, and designer; with a check done to ensure the item ID does not belong to an existing category.
- **Exit:** Exit the program

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Assumptions Made

Keyword search - keyword search is case sensitive and only matches against the exact furniture name where partial substring matches are not supported. The keyword must match the exact full name as stored in the database.

Add furniture - it is assumed that the user will only add furniture item fields that are mandatory, with the rest set to null.

Teamwork & Coordination

The primary method of communication was WhatsApp, with meetings held both online and in person every 3-4 days. Tasks and functions were split across the group according to difficulty and time available.

Member Contributions			
Person	Work Item	Estimated Time Spent	Progress
Maheen			
	connect_db()	15 minutes	Created and completed the connect_db() function to connect to the MongoDB server, given a port. This function is in both python files.
	reset_collection()	15 minutes	Created and completed the function to drop existing furniture collections
	load() and build_doc()	1 hour	Created and completed the functions to (1) read a json file, (2) convert each entry into a doc, and (3) insert the doc into the MongoDB.
	menu()	10 minutes	Created and completed the function to display a basic menu for the furniture db.
	discount_check()	1 hour	Created and completed the function to check if an item is on discount
	Helper functions	1 hour	Created and completed helper functions: paginate_list(); center_print; center_divider()
	main() [both files]	25 minutes	Created and completed the main() functions for operations.p and load_json.py
Harnoor	Keyword search and category search functions in operations.py	5 hrs	Completed keyword and category search queries and added pagination for each
Manaal			
	add_furniture()	50 min	Created and completed add_furniture

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Member Contributions			
Person	Work Item	Estimated Time Spent	Progress
			function
	Report Doc	2.5 hours	Created and completed system overview, user guide, and created template for member contributions